

Algebra II with Trigonometry

Chapter 7.4 & 7.5

Olympiad Solutions (Gemini)

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[4626701_alg-2trig-h-chp-7-4-7-5-note-docx.pdf](#)

Solutions

1. Find the sum of all distinct solutions $x \in [0, 2\pi)$ to the equation $\sin x + \sin 2x + \sin 3x = 0$.

Solution: We begin by applying the sum-to-product identity,
 $\sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$, to the first and third terms:

$$\sin x + \sin 3x = 2 \sin(2x) \cos(x)$$

Substituting this back into the original equation, we can factor out $\sin 2x$:

$$2 \sin(2x) \cos(x) + \sin 2x = 0 \implies \sin 2x(2 \cos x + 1) = 0$$

This gives two cases: 1.

$$\sin 2x = 0 \implies 2x \in \{0, \pi, 2\pi, 3\pi\} \implies x \in \left\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\right\}$$

$$\cos x = -\frac{1}{2} \implies x \in \left\{\frac{2\pi}{3}, \frac{4\pi}{3}\right\}$$

All these solutions are distinct and lie in the interval $[0, 2\pi)$. Their sum is:

$$\left(0 + \frac{\pi}{2} + \pi + \frac{3\pi}{2}\right) + \left(\frac{2\pi}{3} + \frac{4\pi}{3}\right) = 3\pi + 2\pi = 5\pi$$

2. Evaluate the exact numerical value of the following expression without a calculator: $\tan 20^\circ + \tan 40^\circ + \sqrt{3} \tan 20^\circ \tan 40^\circ$

Solution: Observe that $20^\circ + 40^\circ = 60^\circ$. We apply the tangent addition formula, $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$:

$$\tan(60^\circ) = \frac{\tan 20^\circ + \tan 40^\circ}{1 - \tan 20^\circ \tan 40^\circ}$$

Since $\tan 60^\circ = \sqrt{3}$, we have:

$$\sqrt{3} = \frac{\tan 20^\circ + \tan 40^\circ}{1 - \tan 20^\circ \tan 40^\circ}$$

Multiplying both sides by the denominator yields:

$$\sqrt{3} - \sqrt{3} \tan 20^\circ \tan 40^\circ = \tan 20^\circ + \tan 40^\circ$$

Rearranging the terms, we immediately obtain:

$$\tan 20^\circ + \tan 40^\circ + \sqrt{3} \tan 20^\circ \tan 40^\circ = \sqrt{3}$$

3. Given that $\tan x + \cot x = 4$, determine the exact value of $\sin^4 x + \cos^4 x$.

Solution: First, we rewrite the given equation in terms of sine and cosine:

$$\tan x + \cot x = \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{1}{\sin x \cos x}$$

Setting this equal to 4 gives $\sin x \cos x = \frac{1}{4}$.

Now, we can evaluate the target expression by completing the square:

$$\sin^4 x + \cos^4 x = (\sin^2 x + \cos^2 x)^2 - 2 \sin^2 x \cos^2 x$$

Since $\sin^2 x + \cos^2 x = 1$, we substitute our known product:

$$\sin^4 x + \cos^4 x = 1^2 - 2 \left(\frac{1}{4} \right)^2 = 1 - 2 \left(\frac{1}{16} \right) = 1 - \frac{1}{8} = \frac{7}{8}$$

4. Find all real solutions $x \in [0, 2\pi)$ that satisfy the equation $\sin^8 x + \cos^8 x = 1$.

Solution: For any real number x , we know that $0 \leq \sin^2 x \leq 1$ and $0 \leq \cos^2 x \leq 1$. Raising these fractions to higher powers strictly decreases or maintains their value, meaning:

$$\sin^8 x \leq \sin^2 x \quad \text{and} \quad \cos^8 x \leq \cos^2 x$$

Adding these inequalities gives:

$$\sin^8 x + \cos^8 x \leq \sin^2 x + \cos^2 x = 1$$

Equality holds if and only if $\sin^8 x = \sin^2 x$ and $\cos^8 x = \cos^2 x$, which requires both $\sin^2 x \in \{0, 1\}$ and $\cos^2 x \in \{0, 1\}$. This occurs precisely when either $\sin x = 0$ or $\cos x = 0$. In the interval $[0, 2\pi)$, the solutions correspond to the coordinate axes:

$$x \in \left\{ 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2} \right\}$$

5. Evaluate the exact value of $\sec^4\left(\frac{\pi}{12}\right) + \csc^4\left(\frac{\pi}{12}\right)$.

Solution: Let $x = \frac{\pi}{12}$. We rewrite the expression in terms of sine and cosine:

$$\sec^4 x + \csc^4 x = \frac{1}{\cos^4 x} + \frac{1}{\sin^4 x} = \frac{\sin^4 x + \cos^4 x}{\sin^4 x \cos^4 x}$$

Using the identity $\sin^4 x + \cos^4 x = 1 - 2\sin^2 x \cos^2 x = 1 - \frac{1}{2}\sin^2(2x)$, we evaluate the numerator for $2x = \frac{\pi}{6}$:

$$1 - \frac{1}{2}\sin^2\left(\frac{\pi}{6}\right) = 1 - \frac{1}{2}\left(\frac{1}{2}\right)^2 = 1 - \frac{1}{8} = \frac{7}{8}$$

The denominator can be rewritten using the double angle identity $\sin x \cos x = \frac{1}{2}\sin(2x)$:

$$\sin^4 x \cos^4 x = \left(\frac{1}{2}\sin\left(\frac{\pi}{6}\right)\right)^4 = \left(\frac{1}{2} \cdot \frac{1}{2}\right)^4 = \frac{1}{256}$$

Dividing the numerator by the denominator gives the final result:

$$\frac{7/8}{1/256} = \frac{7}{8} \times 256 = 7 \times 32 = 224$$

6. Determine the unique angle $x \in (0, \frac{\pi}{2})$ that satisfies the equation $\cos x + \sin x = \sqrt{2} \cos(\frac{x}{2})$.

Solution: We can consolidate the left-hand side into a single cosine function by multiplying and dividing by $\sqrt{2}$:

$$\cos x + \sin x = \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x \right) = \sqrt{2} \left(\cos x \cos \frac{\pi}{4} + \sin x \sin \frac{\pi}{4} \right)$$

The original equation simplifies to:

$$\sqrt{2} \cos \left(x - \frac{\pi}{4} \right) = \sqrt{2} \cos \left(\frac{x}{2} \right) \implies \cos \left(x - \frac{\pi}{4} \right) = \cos \left(\frac{x}{2} \right)$$

Since $x \in (0, \frac{\pi}{2})$, the arguments reside in bounded intervals:

$x - \frac{\pi}{4} \in (-\frac{\pi}{4}, \frac{\pi}{4})$ and $\frac{x}{2} \in (0, \frac{\pi}{4})$. For cosines of angles in this range to be equal, the arguments must either be equal or be opposites of each other: 1.

$x - \frac{\pi}{4} = \frac{x}{2} \implies \frac{x}{2} = \frac{\pi}{4} \implies x = \frac{\pi}{2}$ (Extraneous, as x is strictly in the open interval $(0, \frac{\pi}{2})$) 2. $x - \frac{\pi}{4} = -\frac{x}{2} \implies \frac{3x}{2} = \frac{\pi}{4} \implies x = \frac{\pi}{6}$

Thus, the unique angle is $x = \frac{\pi}{6}$.